

TG ECET (WP)-2026**SYLLABUS FOR MECHANICAL ENGINEERING****MATHEMATICS (10 Marks)**

S.No	Syllabus
1	UNIT I: MATRICES AND DETERMINANTS Definition of Matrix-order of a matrix-Type of matrices - Algebra of matrices- sum, difference, scalar multiplication and product of 2x2 Matrix - Equality of two matrices-Transpose of a Matrix – Determinant of a square Matrix of order 2×2 .
2	UNIT II: CO-ORDINATE GEOMETRY Straight lines: Slope of straight line - Various forms of straight line: Point-Slope form, Two- Point form, Intercept form, and General equation of a Straight-line Conditions for Parallelism and Perpendicularity of the Straight lines – Equation of a straight line parallel and perpendicular to the given straight line -Distance between two Parallel lines.
3	UNIT III: TRIGONOMETRY Trigonometric Ratios of Compound angles (without proof) - Trigonometric Ratios of Multiple and Sub- multiple angles (2A, 3A, A/2).

MECHANICAL ENGINEERING (90 Marks)

S.No	Syllabus
1	Unit I: BASIC MANUFACTURING TECHNOLOGY Carpentry, Forging and Sheet Metal Tools - Fitting & Drilling - Mechanical Working of Metals - Lathe & Lathe Works – Foundry - Welding
2	Unit II: ADVANCED MANUFACTURING TECHNOLOGY Production Lathes - Cutting Fluids - Advanced Casting methods – Advanced Welding Methods – Milling - Grinding Operations - Finishing Operations - Modern Machining Processes - Press Tools, Jigs & Fixtures
3	Unit III: ENGINEERING MATERIALS Mechanical properties and testing of materials - structure of materials - Production of Iron and Steel – Iron-Carbon equilibrium Diagram - Heat treatment of Steel - Ferrous, Non-Ferrous Metals : their alloys - Powder Metallurgy.
4	Unit IV: CAD/CAM Introduction to Computer Aided Design - Introduction to Computer Aided Manufacturing - NC and CNC Machines - CNC Part Programming - CIM and Introduction to Additive Manufacturing - Fundamentals of Robotics and Automation.

5	Unit V: INDUSTRIAL MANAGEMENT Principles and functions of management, organization structure and behavior, Production and materials management, Marketing sales and Feasibility study, Introduction to ISO 9000, Total Quality Management, Industrial legislation and safety, Entrepreneurial development.
6	Unit VI: THERMODYNAMICS Fundamentals of Thermodynamics - Perfect gas laws- Thermodynamic processes - Fuels and Combustion - Air Standard Cycles – IC engines -Working principles, Systems used and performance.
7	Unit VII: HEAT POWER ENGINEERING Air Compressors, Gas turbines and Jet propulsion. Properties of Steam, Working and Performance of Steam Boilers, Steam nozzles, Steam Turbines and Steam Condensers.
8	Unit VIII: REFRIGERATION AND AIR CONDITIONING Air refrigeration: Reverse Carnot cycle, Bell Coleman cycle - Vapour compression refrigeration system – Vapour absorption refrigeration system – Refrigerants – Refrigeration equipment – Air conditioning - Psychrometry – Air distribution – Applications of refrigeration – Applications of air conditioning.
9	Unit IX: FLUID MECHANICS AND HYDRAULIC MACHINERY Properties of Fluids, Flow of liquids, Flow through pipes, Impact of Jets, Water Turbines - Reciprocating and Centrifugal pumps.
10	Unit X: SOLID MECHANICS System of forces -Simple Stresses and Strains - Geometrical Properties of Sections - Shear Force and Bending Moment diagrams – Theory of Simple Bending - Deflection of Beams - Torsion in shafts.
11	Unit XI: DESIGN OF MACHINE ELEMENTS Design of Bolts, Nuts and Screws – Shafts - Keys and Couplings - Bearings,- Design of Belt and Gear drives -Cams.